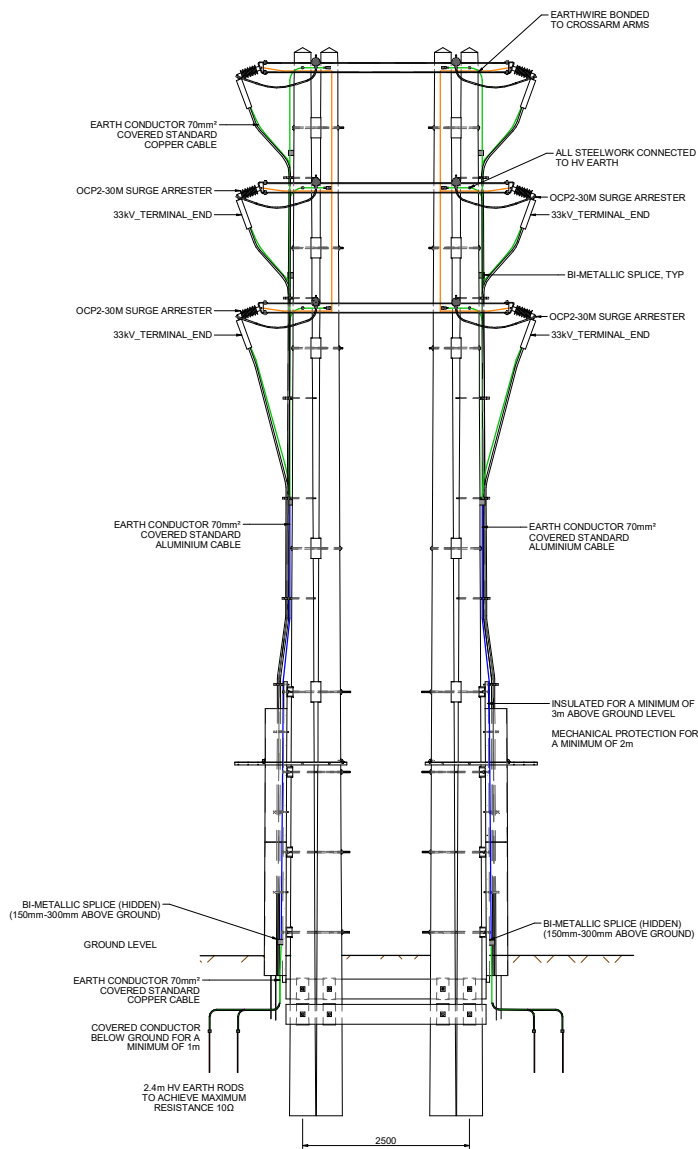
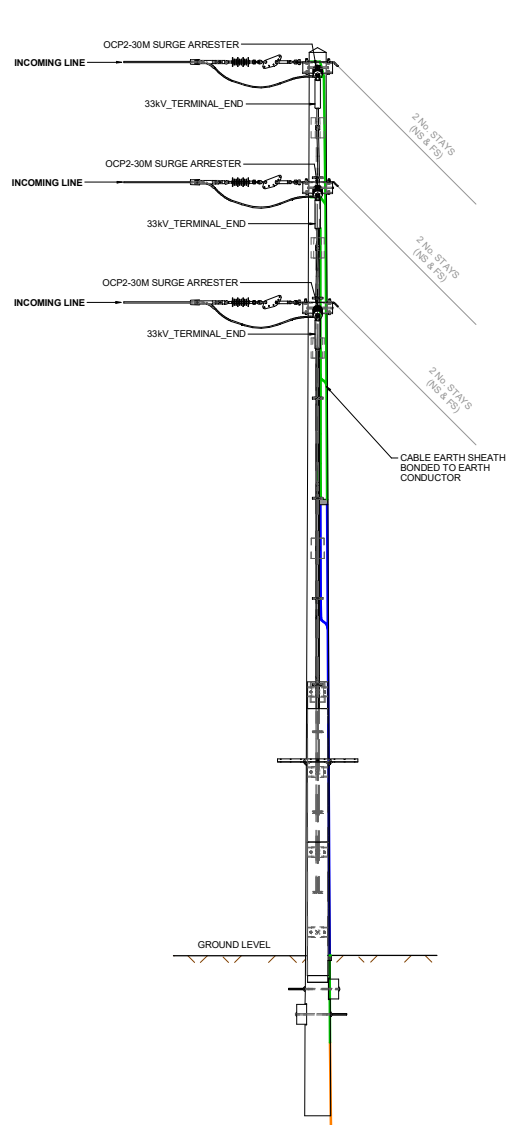




TRANSVERSE ELEVATION



LONGITUDINAL ELEVATION

EARTHING AND BONDING CONDUCTORS			
Function	Network	Fault Level	Earthing and Bonding Conductors ^a
Earth Conductor	EPN	upto 4kA	70mm² covered stranded aluminium cable
	EPN	upto 4kA	35mm² covered stranded copper cable
Bonding Conductor	EPN/SPN	All	35mm² covered stranded aluminium cable
	EPN/SPN	All	16mm² covered stranded copper cable

KEY:	
—	70mm² COVERED COPPER EARTH CONDUCTOR
—	70mm² COVERED ALUMINIUM EARTH CONDUCTOR
—	BARE EARTH CONDUCTOR

REV	DATE	BY	CHKD	APPR	SG
A	14/02/2022	RL	RF		

DESCRIPTION FIRST ISSUE

GENERAL NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES (mm) UNLESS NOTED OTHERWISE.
2. MAIN HV EARTH CONDUCTOR SHALL BE KEPT AS STRAIGHT AS POSSIBLE WITH NO SHARP BENDS.
3. HV EARTH ELECTRODE INSTALLED A MINIMUM OF 1m BELOW GROUND.
4. EARTH CONDUCTOR TO BE INSULATED TO A MINIMUM DEPTH OF 1m BELOW GROUND.
5. MAXIMUM RESISTANCE OF HV EARTH ROD SHALL BE 10Ω.
6. OVERHEAD EARTH CONDUCTOR TO BE CONNECTED TO CABLE SHEATH TO PROVIDE EARTH CONTINUITY BETWEEN SUBSTATIONS.
7. ALL STEELWORK TO BE BONDED TO MAIN HV EARTHING CONDUCTOR.
8. CABLE TO BE CONNECTED TO HV EARTH ELECTRODE.
9. STEELWORK AND EYEBOLTS SHOULD BE BONDED TOGETHER AND CONNECTED TO THE HV EARTH ROD.
10. EARTH RODS 12m COPPER CLAD RODS JOINED TO PROVIDE REQUIRED DEPTH. RODS SPACED AT 3m CENTRES, IF MORE THAN 1 REQUIRED.
11. EARTHING IN ACCORDANCE WITH EDS 06-0015 POLE MOUNTED EQUIPMENT EARTHING DESIGN.
12. SURGE ARRESTER IN ACCORDANCE WITH ADVICE FROM UK POWER NETWORKS.
13. ANY SITE DRILLED HOLES OR MODIFIED STEELWORK TO HAVE TWO COATS OF GALVAPROID OR SIMILAR.
14. TO BE USED IN CONJUNCTION WITH THE FOLLOWING:
GENERAL ARRANGEMENT DRAWING No. 15_210521_07
15. NUMBER OF EARTH RODS TO BE DETERMINED BASED UPON GROUND RESISTIVITY. EARTH ELECTRODE RESISTANCE NOT TO EXCEED 10Ω.
16. DESIGN TO BE REASSESSED IF STRUCTURE IS OFTEN FREQUENTED.

STAYS

6 No. 7 / 4.0 GRADE 1150 STAY WIRES IN ACCORDANCE WITH ENATS 43-91.

TYPE 2 RODS AND BLOCKS - SML 101kN.

TYPE 2 STAY BLOCKS TO BE INSTALLED AT A DEPTH OF 1.8m BELOW GROUND.

SEE MATERIAL LIST No. 19_15048_09.

CDM RESIDUAL RISK

☐ DESIGN BASED HAZARDS ARE ACTIVELY ELIMINATED WHERE PRACTICAL. WHERE HAZARDS ARE NOT ELIMINATED THEY ARE IDENTIFIED BY THIS SYMBOL.

☐ HAZARDS / RISKS THAT SHOULD BE CONSIDERED BY A COMPETENT CONTRACTOR ARE NOT IDENTIFIED.

IF IN DOUBT ASK ENGINEER / DESIGN OFFICE

VIEW CONVENTION
TOWARD ANGLE

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CLIENT
UK Power Networks

PROJECT
BRAINTREE - DUNMOW

TITLE
EARTHING SKETCH FOR 33kV 'H' TERMINAL POLE ARRANGEMENT

SCALE 1:30	DESIGNED LSTC
DATE 11/02/2022	CHECKED RF
DRAWN RL	APPROVED SG

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DESIGN	DRAWING NUMBER	REV
A1	23_210521_11	A